

Immunopathology Basics - Factoid Sheet

Week 8 • Foundations of Microbiology I • Dolen current deck controls

I = IgE immediate • II = antibody on fixed target • III = soluble immune complexes deposit • IV = T cells delayed.

Gell & Coombs

Type	Mechanism	Deck examples	Fast cue
I	IgE, FcεRI, mast-cell activation	Rhinitis, asthma, atopic dermatitis, anaphylaxis	Immediate
II	IgG/IgM against cell-surface antigen or receptor	Drug hemolysis, Rh disease, Graves, MG	Fixed target
III	IgG/IgM soluble immune complexes	Serum sickness, Arthus, HP	Deposit elsewhere
IV	T-cell mediated, usually Th1; can include Tc killing	Contact dermatitis, tuberculin response	1-3 days

Faculty caveat: mechanisms overlap clinically. Exams ask for the **predominant** mechanism.

Allergy / atopy

- **Allergy** in the deck's preferred US usage = IgE-dependent Type I mechanism, **not** a disease/diagnosis.
- Classic "atopic diseases" = rhinitis + asthma + atopic dermatitis; inherited together ("atopic march").
- Non-IgE biology exists: epithelial **TSLP** → **ILC2**; tezepelumab targets TSLP.
- Chronic urticaria is very rarely IgE-dependent; FcεRI autoantibody can activate mast cells.

Type I trigger

IgE-FcεRI waits → allergen cross-links IgE → receptor aggregation → mast-cell activation → histamine/lipid mediators/cytokines → symptoms + eosinophil recruitment.

Tryptase

Clinically useful mast-cell activation biomarker. Deck: basophils do **not** have tryptase.

BTK figure

Conceptual outputs: granule exocytosis + new lipid mediators + cytokine production.

Eosinophil products

Preformed	After activation
Peroxidase; collagenase; major basic protein; eosinophil cationic protein; eosinophil-derived neurotoxin	IL-3, IL-5, GM-CSF; CXCL8; LTC4/LTD4/LTE4; platelet-activating factor

Early vs late

Early Type I: minutes, preformed mast-cell mediators.

Late Type I: hours, inflammatory response.

Do NOT confuse late-phase Type I with Type IV DTH.

Types II-IV + Therapy

Type II - fixed antigen/receptor

Pattern	Deck examples
Cytotoxic	Penicillin-induced hemolytic anemia; Rh hemolytic disease. Complement → lysis and/or enhanced phagocytosis.
Stimulatory	Graves disease.
Inhibitory	Autoimmune hypothyroidism; myasthenia gravis.
FcεRI autoantibody	Autoimmune chronic urticaria via non-IgE mast-cell activation.

II vs III: II antibody binds the target directly. III complexes form in circulation first.

Type III - immune complexes

Pattern	Recognition
Arthus	Local; preformed IgG; local IC deposition/complement. Example: sore arm hours after dT booster.
Serum sickness	Systemic; foreign antigen; rash + GN + arthritis + fever as antibody forms, usually ~1 week.
Hypersensitivity pneumonitis	Inhaled antigen → IgG → IC deposition in alveoli; farmer's lung / bird fancier's lung.

Common deposition outcomes: glomerulus → nephritis; synovium → arthritis; postcapillary venules → vasculitis.

Type IV - T cells

- Usually **1-3 days** after antigen contact.
- Th1 + macrophage inflammatory pathway; processed antigen presented by MHC II.
- Can also be cytotoxic T-cell mediated.
- DTH = detrimental delayed hypersensitivity; CMI = beneficial cell-mediated immunity.
- Examples: contact dermatitis, transplant rejection, fungal resistance, Hashimoto thyroiditis.

Therapy slide 46

Strategy	Deck examples
Broad/pathway suppression	Corticosteroids; JAK inhibitors; -brutinibs
Monoclonal antibodies	Omalizumab; benralizumab; tezepelumab
Mediator blockade	H1 antihistamines; leukotriene-receptor antagonists
Allergen immunotherapy	Allergy shots for IgE-proven rhinitis, asthma, selected conditions

Companion-only enrichment: ACID mnemonic, Th2 IL-4/IL-13/IL-5 sensitization detail, allergy testing, epinephrine pharmacology, and broader DMARD/autoimmunity discussion are in the Word guide but are not dedicated current-deck content.

Final recognition: IgE/mast cell = I • fixed Ab target = II • traveling complexes = III • T cell + delay = IV.

Primary: current 49-slide Dolen deck. Word companion is secondary enrichment only.